Title:

Create a small web-based Gym Management System by selecting relevant system environments, platforms and programming languages. Prepare a concise test plan with features to be tested and bug taxonomy, regression scripts, and test reports using Selenium WebDriver and IDE including exploratory testing.

Problem Definition:

Perform web testing, identify bugs using Selenium WebDriver and Selenium IDE, and generate test reports for the Gym Management System.

Prepared By:

Name: _____ Roll No.: _____
Class: BE Computer Engineering Division: _____
College: _____ Faculty: _____

Prerequisite:

Knowledge of Core Java.

Software Requirements:

Software	**Requirement / use**
Operating system	Windows; the project uses Windows PowerShell scripts.
Java	JDK 17 or newer; tested with JDK 17.0.12. Java HTTP server and JDBC backend.
Database	MySQL 8.x; dedicated project instance at 127.0.0.1:3307.
Web interface	HTML, CSS and JavaScript; Chrome or Edge. App URL: http://127.0.0.1:8081.
Testing tools	Selenium WebDriver 4.51.0 (Java and Python), Selenium IDE runner 4.0.13; Python 3.10+ and Node.js for the supplied runners.

Hardware Requirement:

Suggested local demonstration configuration: a dual-core processor, 8 GB RAM, sufficient free storage for the JDK, MySQL and browser, a keyboard, mouse and display. These are planning recommendations rather than measured minimum system requirements.

Theory:

Introduction of mini project

The Gym Management System maintains member and trainer profiles, memberships, attendance, equipment, products, fitness measurements, account permissions and related histories. The browser edition reproduces the existing Java Swing application using Java 17, JDBC and the same MySQL schema. HTML, CSS and JavaScript provide the browser interface.

Admin manages records, account access and membership prices, views payment and sales histories, downloads six operational PDF reports and creates a local SQL backup. Trainer views assigned members and records progress for an active assigned member. Member views their own profile, membership, attendance, progress and training history. Trainer and Member can change their permitted profile fields and password.

The account role and linked profile determine access after sign-in. Passwords use salted PBKDF2 hashes, JDBC statements use parameters, and related changes use database transactions. Payment and sales entry, workout editing and creation of training assignments are outside the inherited interface; their stored histories and detail views remain available.

Testing Tool: Selenium

Selenium automates interactions with a web application in a real browser. Test scripts locate page elements, enter data, click controls, wait for page changes and assert expected results. It supports repeatable checks of authentication, forms, navigation, role access and downloads.

Selenium WebDriver controls a browser through its driver. Selenium IDE stores command sequences in a .side project that can be replayed. Selenium Grid coordinates browser sessions across machines. Selenium Remote Control (RC), named in the supplied theory format, is a legacy component; this project uses WebDriver and IDE replay.

The detailed regression is written in Python WebDriver and executed on Chrome and Edge. An additional Java WebDriver program demonstrates Core Java test code. The 30-case Selenium IDE project was executed by the official command-line replay engine. Graphical IDE recording/playback was not used as execution evidence.

Advantages of Selenium

i) Open-source browser automation.
ii) Supports multiple browsers and programming languages.
iii) Repeats regression checks consistently.
iv) Supports explicit waits and browser sessions.
v) Can be integrated with test runners and result-processing scripts.

Disadvantages of Selenium

i) Requires a supported browser and a compatible driver.
ii) Changing page structure can break element locators.
iii) Timing and asynchronous page changes require suitable waits.
iv) Image and visual layout checks require additional inspection.
v) Reports and test data cleanup must be provided by the test framework or supporting scripts.

Test Plan

INTRODUCTION

This test plan defines the scope, approach, resources, responsibilities, schedule and criteria for testing the Gym Management System web application. It covers the Admin, Trainer and Member workflows implemented in the current project.

1 TEST STRATEGY

1.1 Scope of Testing

1.1.1 Feature to be tested

Module Name	Applicable Roles	Description
Login and logout	All roles	Valid username/mobile, incorrect input, inactive accounts and session logout.
Dashboard	All roles	Saved-record totals, own-role metrics and recorded weekly activity.
Member management	Admin	Create with initial measurements/BMI, search, edit, membership/payment flags, activate/deactivate and delete.
Trainer management	Admin	Create trainer and linked login together; edit/status/delete and validation.
Attendance	Admin; scoped views for Trainer/Member	Active member check-in, duplicate prevention, payment-due warning and own/assigned history.
Equipment and products	Admin	Create/edit/delete, stock, condition, numeric bounds and referenced-record protection.
Memberships and workouts	Admin	Plan-price validation/update and readable seven-day workout details.
User management	Admin	Linked roles, unique profiles, active state, password reset and self-protection.
Progress and training	Trainer; Member reads own data	Assigned-member progress, calculated BMI, date/measurement bounds and scope restrictions.
Profile and password	Trainer, Member	Own profile changes, current-password verification and confirmation validation.
History and exports	Admin	Payment/sales histories, filtered CSV, six PDF downloads and local SQL backup.
Interface and access	All roles	Search, pagination, mobile navigation, keyboard focus, theme and protected routes.

1.1.2 Feature not to be tested

Production deployment, load/performance benchmarking, a comprehensive penetration test, hardware interfaces and the original Swing interface are outside this web test cycle. New payment/sale entry and editing workout or training assignments are outside the current application workflow. Existing related history/detail views are tested.

1.2 Test Type

Test type	Application in this project
Integration testing	Java HTTP routes, JDBC and MySQL persistence; paired member/progress and trainer/account saves.
System testing	Complete sign-in, role workflow, record management and export flows through the browser.
API testing	Authenticated browser fetch requests verify server-side bounds, forbidden roles and invalid/duplicate actions.
Regression testing	79 detailed scenarios on each browser, 30 IDE replay scenarios and 8 Java smoke scenarios.
Compatibility and UI	Chrome/Edge runs, 1440px desktop and 390px mobile layouts, keyboard and theme behavior.
Exploratory testing	Independent role sessions, modal focus/Escape, mobile outside click and measurement boundaries.

Regression scripts

Detailed suite: web/tests/selenium_suite.py, launched by Verify-Web.ps1.
Java companion: web/tests/java/src/main/java/GymWebSmoke.java, launched by Verify-WebJava.ps1.
IDE project: web/tests/Gym_STQA.side, launched by Run-SeleniumIDE.ps1.
Exploration: web/tests/exploratory_checks.py.
Scripts use stable identifiers and explicit waits. Detailed suites create uniquely tagged test records, verify the database data-directory path and clean up only their fixture IDs.

1.3 Risk and Issues

Risk	Mitigation
An unrelated service occupies a project port.	Verify MySQL 3307 ownership before setup. Stop setup for a foreign listener. Use the documented free web port 8081.
Browser and driver versions differ.	Use Selenium Manager or a matching cached ChromeDriver; record the actual browser version in results.
Dynamic forms and page transitions affect selectors.	Scope locators to the visible dialog, use stable IDs, explicit waits and mobile device emulation.
Tests accidentally alter existing gym data.	Verify this checkout database; create distinct fixtures, use exact IDs and preserve original profiles/passwords.
PDF compatibility or overflowing rows.	Use OpenPDF first in the classpath, parse downloaded PDFs and visually inspect wrapping/page breaks.

Bug taxonomy and identified defects

The following observations were recorded during the migration and test work. Discovery by code inspection is identified explicitly; those observations are not represented as Selenium-discovered failures.

ID / category	Observation / discovery	Correction and evidence
D01 Database compatibility	Live request: an unquoted member SQL alias caused dashboard HTTP 500.	Quoted aliases; dashboard and module regressions pass.

ID / category	Observation / discovery	Correction and evidence
D02 Report library compatibility	Live PDF request: legacy iText/OpenPDF conflict caused NoSuchMethodError.	OpenPDF precedence; TC42-TC47 download/parse all six PDFs.
D03 Startup environment	Actual launch: selector socket setup failed; web port 8080 was occupied.	Project-local socket directory and web port 8081; browser connection succeeds.
D04 Functional/data integrity	Inspection: draft lacked atomic initial measurements and trainer-account creation.	Transactional paired saves and valid form IDs; TC14, TC15 and TC23 pass.
D05 Validation	Inspection: some numeric bounds and password confirmation checks were incomplete.	Server and form validation; negative value/date/password cases pass.
D06 Automation compatibility	Selenium: select_by_value could not find implicit page-size option values.	Explicit 10/25/50 values; pagination and TC79 pass.
D07 Report layout	Visual inspection: DAYPASS wrapped into an adjacent member-report row.	Automatic table-row heights and readable PDF layouts; downloads were rerun.
D08 Keyboard interface	Focused Selenium check: Tab moved focus outside the open dialog.	Explicit first/last-control wrapping; 12 Tab transitions and Escape pass.

Test-harness issues included a hidden-field locator, mobile transition timing, and Windows path quoting in the IDE runner. Their diagnostic failures were corrected and retained separately. Final acceptance uses the passing runs identified in the execution summary.

1.4 Test Logistics

1.4.1 Who will test?

The student/team named under Prepared By owns test planning, script review, exploratory review and the faculty demonstration. Automated WebDriver and IDE scripts execute repeatable checks. Student names and roll numbers must be completed before submission.

1.4.2 When will test occur?

Testing starts when the application and dedicated database are available, specifications and test data are ready, browser drivers are available, and the environment ownership checks pass. The recorded final runs were executed on 15 October 2026 after corrections.

2 TEST OBJECTIVE

Verify correct authentication and role access; reliable member/trainer and inventory management; persistent attendance and progress; validation and relationship protection; readable reports/exports; and an operable desktop/mobile interface. Confirm that test fixtures are removed without resetting existing profiles or passwords.

3 TEST CRITERIA

3.1 Suspension Criteria

Suspend when the database/server is unavailable, a foreign database listener is detected, test-data isolation cannot be established, or 40% of the current cases fail as described in the supplied plan example. Repair the blocking cause and rerun before acceptance.

3.2 Exit Criteria

Run rate must be 100%. The supplied format gives an 80% pass-rate threshold; this project uses the stricter requirement of zero final failed cases in the planned suites, successful fixture cleanup, and complete saved reports/screenshots. The documented final runs meet these criteria.

4 RESOURCE PLANNING

4.1 System Resource

No.	Resource	Description
1	Application server	Java HTTP server on this computer at http://127.0.0.1:8081.
2	Database server	This checkout MySQL instance at 127.0.0.1:3307; schema gms.
3	Test tools	Selenium 4.51.0, Chrome 155.0.8059.79, Edge 155.0.4283.45, IDE runner 4.0.13.
4	Network	Local loopback during demonstration; Internet is needed only for missing initial dependencies/drivers.
5	Computer	Windows machine with JDK, MySQL binaries and an installed supported browser.

4.2 Human Resource

Role	Responsibility
Project student/team	Review the requirements and project, prepare test data, execute scripts, inspect failures, verify fixes and maintain the report.
Faculty reviewer	Review the submitted report and observe the project/test demonstration.

5 SCHEDULE & ESTIMATION

5.1 All project task and estimation

The effort values below are planning allowances for a local test/report cycle, not measured labour hours or claims about actual elapsed implementation time.

Task	Member / role	Estimated effort
Review requirements and features	Project student/team	30 minutes
Prepare cases and scripts	Project student/team	45 minutes
Execute tests and investigate defects	Project student/team	90 minutes
Review fixes and rerun regression	Project student/team	45 minutes
Prepare report and demonstration	Project student/team	30 minutes
Total planning allowance	Project student/team	240 minutes

Schedule to complete these task

Stage	Sequence / completion
Before testing	Inspect assignment/project; prepare environment, plan, cases and test data.
During testing	Run WebDriver, correct observed issues, repeat Chrome/Edge, replay IDE and run focused/Java checks.
After testing	Confirm final passing JSON and cleanup, capture screenshots, prepare report, package and demo instructions.

TEST DELIVERABLES

7.1 Before testing phase

- Test plan and feature scope.
- Test cases and procedures.
- Test design and data-isolation rules.

7.2 During the testing

- WebDriver/IDE/Java scripts and test data.
- Browser execution traces, error records and screenshots.
- Requirements traceability matrix (requirements_traceability.csv).
- Local JSON, CSV and HTML results; separate diagnostic observations.

7.3 After the testing cycles is over

- Final test results and this report.
- Defect report and exploratory results.
- Project run and test instructions.
- Source/report submission archive and release notes in web/README.md.

Test Report for Test Cases inclusive of Test Procedures for identified Test Scenarios.

All 79 cases below were executed independently on Chrome and Edge. The Actual Result column records the final Chrome observation; Pass means the corresponding case passed on both browsers. The separate raw files preserve each browser result and fixture identifier.

Sr. No.	Test Objective	Steps	Expected Result	Actual Result	Test Status
TC01	Required sign-in fields	Open the sign-in page. Submit with both fields empty. Check the required-field validation.	Blank form stays on the sign-in page	Required-field validation blocked the empty sign-in form.	Pass
TC02	Unknown account rejected	Enter an unknown username and a password. Submit and inspect the error and dashboard visibility.	Error shown; no authenticated dashboard	Error shown; no authenticated dashboard	Pass
TC03	Incorrect password rejected	Enter admin with an incorrect password. Submit and inspect the error.	Error shown for a valid account with wrong password	Error shown for a valid account with wrong password	Pass
TC04	SQL injection input rejected	Enter an SQL-injection-style username and a password. Submit and check that authentication is denied.	Injected username does not authenticate	Injected username does not authenticate	Pass
TC05	Admin username login	Enter admin / 123. Submit and wait for the Admin overview.	Admin overview loads with real metrics	Admin overview loads with real metrics	Pass
TC06	Admin dashboard metrics	Inspect the five overview cards. Read dashboard data through the authenticated browser request and compare loaded metrics.	Five cards loaded with database totals including product sales	Five real metric cards and weekly attendance chart loaded.	Pass
TC07	Admin mobile-number login	Sign in with mobile 0770000001 and password 123. Confirm the Admin workspace.	Mobile number and password authenticate admin	Mobile number and password authenticate admin	Pass
TC08	Member search	Open Members. Search for an existing member name. Inspect the matching rows.	Matching member rows remain visible	Matching name: Diya Banerjee	Pass
TC09	Search empty state	Search Members for a unique nonexistent string. Inspect the empty-state message.	Unknown search shows No matching records	Empty state shown for unmatched search.	Pass
TC10	Member table pagination	Open Members with 10 rows per page. Select Next and inspect the page indicator and displayed range.	Next page changes visible rows and page indicator	11-20 of 25 records	Pass
TC11	Member required-field validation	Open Add member. Submit without required values. Confirm that the dialog stays open.	Incomplete member form cannot be submitted	Incomplete member form cannot be submitted	Pass

Sr. No.	Test Objective	Steps	Expected Result	Actual Result	Test Status
TC12	Mobile number validation	Enter a three-digit mobile in Add member. Inspect the input pattern validation.	Three-digit mobile is rejected by the form	Three-digit mobile is rejected by the form	Pass
TC13	Initial BMI preview	Enter height 175 cm and weight 72 kg. Read the calculated BMI preview.	175 cm and 72 kg calculate BMI 23.51	BMI preview: 23.51	Pass
TC14	Create member with measurements	Create an isolated member with a unique mobile, Monthly plan, paid flag, health notes, height 175 cm, weight 72 kg, and body fat 18%.	Member and initial progress are saved together	Created member #150	Pass
TC15	Member initial progress persistence	Read Progress tracking for the created member. Verify one initial entry and BMI 23.51.	Initial measurement is available in Progress tracking	Initial measurement found with BMI 23.51.	Pass
TC16	Duplicate member mobile blocked	Try to create another member with the same mobile and complete measurements. Inspect the duplicate error and count matching members.	Duplicate mobile rejected without duplicate records	Duplicate mobile rejected without duplicate records	Pass
TC17	Edit member and retain changes	Edit the fixture member name and payment flag to Due. Save, refresh the browser, and search for the fixture.	Updated name/payment flag survive a browser refresh	Updated name/payment flag survive a browser refresh	Pass
TC18	Deactivate member	Deactivate the fixture member. Confirm the new status and absence from active check-in choices.	Inactive member excluded from check-in choices	Inactive member excluded from check-in choices	Pass
TC19	Reactivate member	Reactivate the fixture member. Confirm Active status.	Member status returns to Active	Member status returns to Active	Pass
TC20	Record attendance with payment warning	Select the fixture member in Record check-in. Save and inspect the payment-due warning and attendance history.	One check-in saved and payment-due warning shown	One check-in saved and payment-due warning shown	Pass
TC21	Duplicate attendance blocked	Attempt a second check-in for the same member on the same date. Inspect the validation error.	Second check-in on same day is rejected	Second check-in on same day is rejected	Pass
TC22	Unknown member attendance rejected	Submit an attendance action with a nonexistent member ID through the authenticated browser. Inspect HTTP 400.	Tampered member ID produces HTTP 400	Choose an active member.	Pass

Sr. No.	Test Objective	Steps	Expected Result	Actual Result	Test Status
TC23	Create trainer and login account	Create a unique trainer with valid profile details, hire date, username, and temporary password. Inspect the trainer and linked account.	Profile and Trainer account created atomically	Trainer and linked login created.	Pass
TC24	Trainer username validation	Enter a trainer username starting with a digit. Inspect the username pattern validation.	Username beginning with a digit is invalid	Username beginning with a digit is invalid	Pass
TC25	Create linked Member account	Create a Member account and link it to the fixture member. Confirm the account appears in User management.	Account links to the selected member profile	Account links to the selected member profile	Pass
TC26	Duplicate profile link rejected	Attempt to link another account to the same member profile. Inspect HTTP 409 and the duplicate-link error.	Second account linked to same member gets HTTP 409	That username, mobile number, profile link, or serial number is already in use.	Pass
TC27	Inactive account cannot log in	Deactivate the fixture Member account. Try to sign in with it and verify the inactive-account error.	Deactivated Member login is rejected	Deactivated Member login is rejected	Pass
TC28	Account reactivation	Reactivate the fixture Member account in User management. Read the updated Active status.	Member account is Active after reactivation	Member account is Active after reactivation	Pass
TC29	Current admin protected	Submit a request to deactivate the current Admin account. Inspect the self-protection error.	Admin cannot disable own account (HTTP 400)	You cannot disable your own account.	Pass
TC30	Create equipment	Create equipment with a unique serial, purchase date, and warranty date. Read the new inventory row.	Equipment profile appears in inventory	Equipment profile appears in inventory	Pass
TC31	Edit equipment condition	Edit the fixture equipment condition to UNDER MAINTENANCE. Save and reread its condition.	Condition is persisted as UNDER MAINTENANCE	Condition is persisted as UNDER MAINTENANCE	Pass
TC32	Delete unreferenced equipment	Delete the unreferenced fixture equipment and confirm it no longer appears.	Created equipment is removed	Created equipment is removed	Pass
TC33	Create product	Create a unique product with price 499.50 and stock 12. Confirm its inventory record.	Product price and stock saved	Product price and stock saved	Pass
TC34	Negative product price validation	Enter -1 as the product price. Inspect the browser's minimum-value validation.	Negative price rejected by browser validation	Negative price rejected by browser validation	Pass

Sr. No.	Test Objective	Steps	Expected Result	Actual Result	Test Status
TC35	Edit product price and stock	Edit the fixture product to price 549.75 and stock 20. Save and inspect persisted values.	Updated price and quantity persist	Updated price and quantity persist	Pass
TC36	Referenced product deletion protected	Add a test-only order reference to the fixture product. Attempt deletion and inspect the protected-history error.	Product with order history cannot be deleted	Product with order history cannot be deleted	Pass
TC37	Delete unreferenced product	Remove the fixture order reference. Delete the product and confirm removal.	Product removed after fixture history is cleared	Product removed after fixture history is cleared	Pass
TC38	Negative plan price rejected	Submit a negative membership-plan price through the browser request. Confirm HTTP 400 without changing the price.	Server rejects negative price (HTTP 400)	Enter a nonnegative amount within the supported range.	Pass
TC39	Workout plan details	Open a workout plan with View plan. Count and read all seven day sections.	All seven workout days are readable	All seven workout days are readable	Pass
TC40	Payment history view	Open Payment history and inspect the loaded records.	Saved payment records loaded in browser	Payment history loaded with 19 records.	Pass
TC41	Product sales view	Open Product sales and inspect the loaded records.	Saved sale records loaded in browser	Product sales loaded with 20 records.	Pass
TC42	Members PDF report	Open Reports. Download the Members PDF. Parse it and confirm at least one page and a nonempty file.	Valid PDF downloaded through report button	4135 bytes; 1 PDF pages.	Pass
TC43	Trainers PDF report	Download the Trainers PDF through its report button. Parse and inspect the downloaded file.	Valid PDF downloaded through report button	2308 bytes; 1 PDF pages.	Pass
TC44	Attendance PDF report	Download the Attendance PDF through its report button. Parse and inspect the downloaded file.	Valid PDF downloaded through report button	28634 bytes; 11 PDF pages.	Pass
TC45	Payments PDF report	Download the Payments PDF through its report button. Parse and inspect the downloaded file.	Valid PDF downloaded through report button	3300 bytes; 1 PDF pages.	Pass
TC46	Sales PDF report	Download the Product sales PDF through its report button. Parse and inspect the downloaded file.	Valid PDF downloaded through report button	4999 bytes; 2 PDF pages.	Pass

Sr. No.	Test Objective	Steps	Expected Result	Actual Result	Test Status
TC47	Equipment PDF report	Download the Equipment PDF through its report button. Parse and inspect the downloaded file.	Valid PDF downloaded through report button	4483 bytes; 2 PDF pages.	Pass
TC48	Filtered CSV report export	Search Members for the fixture. Select Export CSV. Read the file and verify the matching member is included.	CSV download contains visible matching member	Filtered member CSV contains fixture row.	Pass
TC49	Local database backup	Open Database backup and create a snapshot. Verify the reported SQL file exists inside the checkout's backups folder.	Timestamped SQL backup exists in checkout backups folder	gms_web_20261015_131137 .sql 47275 bytes	Pass
TC50	Trainer sign-in	Create an isolated training assignment and sign in as the fixture trainer. Wait for the Trainer overview.	Trainer account opens a separate workspace	Trainer account opens a separate workspace	Pass
TC51	Trainer navigation restriction	Inspect Trainer navigation. Confirm Assigned members exists and User management does not.	Admin management links are absent	Trainer navigation excludes admin modules.	Pass
TC52	Trainer server authorization	Request the admin Members endpoint from a Trainer session. Inspect HTTP 403.	Direct members API denied with HTTP 403	Administrator access required.	Pass
TC53	Trainer assigned-member list	Open Assigned members. Confirm the fixture member name and ID appear.	Trainer sees their assigned member	Assigned fixture member and ID displayed.	Pass
TC54	Trainer progress entry	Record progress for the assigned member: height 175 cm, weight 71 kg, body fat 17%, current date. Inspect BMI 23.18.	Assigned member progress saved with calculated BMI 23.18	Assigned member progress saved with calculated BMI 23.18	Pass
TC55	Future progress date rejected	Submit a future-dated measurement through the browser request. Inspect HTTP 400.	Server rejects a future measurement date (HTTP 400)	Date cannot be in the future.	Pass
TC56	Unassigned member progress denied	Attempt progress entry for an unrelated member ID. Inspect HTTP 403.	Trainer cannot write progress for an unrelated member	Member is not actively assigned to you.	Pass
TC57	Trainer edits own profile	Open the Trainer profile. Edit the fixture trainer name. Save and inspect the profile and account display.	Profile and account display name update together	Profile and account display name update together	Pass

Sr. No.	Test Objective	Steps	Expected Result	Actual Result	Test Status
TC58	Wrong current password rejected	Attempt a password change with an incorrect current password. Inspect the error.	Password is preserved when current password is wrong	Password is preserved when current password is wrong	Pass
TC59	Password confirmation validation	Enter different new and confirmation passwords. Submit and inspect the mismatch error.	Mismatched new passwords cannot be saved	Mismatched new passwords cannot be saved	Pass
TC60	Trainer password change	Change the fixture trainer password with matching confirmation. Verify that the old password fails and the new password authenticates.	Old password rejected and new password works	Old password rejected and new password works	Pass
TC61	Member sign-in	Sign in with the linked fixture Member account. Wait for the Member overview.	Linked Member account opens member workspace	Linked Member account opens member workspace	Pass
TC62	Member own membership view	Open My profile. Confirm the member's own name, Monthly plan, and payment status.	Member sees own profile, plan, and payment status	Member sees own profile, plan, and payment status	Pass
TC63	Member attendance scoped	Open My attendance. Verify only the fixture member's one check-in is returned.	Only fixture member check-in is returned	Only fixture member check-in is returned	Pass
TC64	Member progress scoped and read-only	Open My progress. Verify the initial and trainer-entered measurements; confirm no add-record action exists.	Own initial/trainer entries visible; no record action	Own initial/trainer entries visible; no record action	Pass
TC65	Member training scoped	Open My training. Confirm only the fixture member's assigned trainer and training record are shown.	Only own training assignment is visible	Only own training assignment is visible	Pass
TC66	Member admin API access denied	Request User management data from the Member browser session. Inspect HTTP 403.	User management API denied with HTTP 403	Administrator access required.	Pass
TC67	Member progress write denied	Attempt a Trainer progress action from the Member session. Inspect HTTP 403.	Member cannot submit trainer progress action	Trainer access required.	Pass
TC68	Member edits own profile	Edit the fixture member's own name. Save and inspect the updated profile.	Own name updates in profile and account	Own name updates in profile and account	Pass
TC69	Desktop layout	Set a desktop browser window to 1440 pixels wide. Inspect the profile and check document horizontal overflow.	Document has no horizontal overflow at 1440px	Document has no horizontal overflow at 1440px	Pass

Sr. No.	Test Objective	Steps	Expected Result	Actual Result	Test Status
TC70	Mobile navigation and layout	Emulate a true 390 by 844 mobile viewport. Open the navigation drawer, select Progress, wait for the transition, and inspect overflow.	390px navigation works without document overflow	390px navigation works without document overflow	Pass
TC71	Theme persistence	Toggle dark theme, refresh the page, and confirm the theme persists. Capture the rendered screen.	Dark theme remains selected after page refresh	Dark theme remains selected after page refresh	Pass
TC72	Sign-out invalidates browser session	Sign out using the confirmation dialog. Confirm the login page and HTTP 401 from the protected account endpoint.	Protected API requires sign-in after logout	Protected API requires sign-in after logout	Pass
TC73	Anonymous report access denied	Request a PDF report without an authenticated session. Inspect HTTP 401.	Report endpoint returns HTTP 401	Your session has ended. Please sign in again.	Pass
TC74	Server source is not publicly served	Request GymWebServer.java through the browser URL. Confirm HTTP 404.	Source file URL returns HTTP 404	Java source URL returned 404.	Pass
TC75	Admin account password reset	As Admin, reset the fixture Member password. Verify the old password fails and the reset password works.	Reset password works; old password no longer works	Reset password works; old password no longer works	Pass
TC76	Delete member and linked account	Delete the fixture member. Verify the profile, linked login, attendance, progress, and assignment are removed and sign-in fails.	Member deletion cascades fixture history and login	Member deletion cascades fixture history and login	Pass
TC77	Delete unreferenced trainer	Delete the now-unreferenced fixture trainer. Confirm profile and linked account removal.	Trainer and linked account removed after assignment cleanup	Trainer and linked account removed after assignment cleanup	Pass
TC78	Module summary values	Compare the Members summary card count with the current authenticated Members response.	Member summary count equals loaded database rows	Module summary matches the actual member record count.	Pass
TC79	Membership plan price update	Create a test-only membership plan. Edit its price from 17.50 to 29.75 in the UI, reread the value, then remove that exact fixture plan.	A valid price change is persisted for an isolated plan	Fixture plan price changed from 17.50 to 29.75.	Pass

Test execution summary

Suite	Pass	Fail	Recorded execution
WebDriver - Chrome	79	0	2026-10-15T13:12:20.393016+05:30
WebDriver - Edge	79	0	2026-10-15T13:14:01.833772+05:30
Selenium IDE replay	30	0	Official runner 4.0.13; raw JSON saved.
Java WebDriver smoke checks	8	0	2026-10-15T13:15:01.527712900+05:30
Focused exploratory checks	8	0	2026-10-15T13:14:29.649529+05:30

204 test executions passed. This comprises 79 detailed scenarios repeated on two browsers, plus overlapping IDE, Java and focused checks. The suites share functional coverage. Fixture cleanup completed without an error.

Exploratory testing report

Additional exploration covered independent role sessions, keyboard interaction, mobile dismissal and invalid measurement boundaries. The final observed results follow.

Test / condition	Actual result	Test Status
Independent Admin and Member browser sessions	Two simultaneous browser sessions retained different roles.	Pass
Dashboard totals match the desktop app	Dashboard includes all members/trainers, matching the desktop COUNT(*) totals.	Pass
Keyboard focus stays in dialog	Twelve Tab transitions stayed inside the native modal.	Pass
Dialog Escape dismissal	Escape closed the unsaved member form.	Pass
Mobile drawer outside-click dismissal	Outside pointer click closed the mobile drawer.	Pass
Height below minimum rejected	height must be between 50.0 and 250.0.	Pass
Body fat above maximum rejected	fat must be between 1.0 and 70.0.	Pass
Non-finite weight rejected	Enter a valid number for weight.	Pass

Long workout text, empty searches, profile refresh, light/dark theme behavior and PDF row wrapping were also reviewed during the recorded work. The first modal-focus diagnostic found focus outside the dialog; the corrected keyboard wrap passed the final 12-Tab test. Downloaded PDFs were parsed and the report layouts were visually inspected.

Screenshot of Code and Output

```
GymWebSmoke.java - actual source excerpt

    static void login(String username, boolean valid) {
        browser.manage().deleteAllCookies(); browser.get(base+"/");

wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("login")));
        element("#login-username").sendKeys(username); element("#login-
password").sendKeys("123"); click("#sign-in");
        if (valid) {

wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("app")));
            wait.until(d -> !d.findElements(By.cssSelector(".stat-
card")).isEmpty());
        } else { wait.until(d -> !element("#login-
error").getText().isBlank()); }
    }

test("JAVA07","Member profile and scoped
navigation",()->{login("user1",true);require(element("#nav-
role").getText().equalsIgnoreCase("member"),"Wrong
role");require(browser.findElements(By.cssSelector("[data-page=users],[data-
page=reports]")).isEmpty(),"Admin links
exposed");page("profile");require(!element("#page-
content").getText().isBlank(),"Empty profile");return "Member profile loads;
Admin navigation absent.";});
```

Figure 1. Actual Java WebDriver login method and Member role test using user1 / 123.

```
Saved Java WebDriver execution results
Source: output/stqa/selenium_java_results.json

Timestamp: 2026-10-15T13:15:01.527712900+05:30
Browser: Chrome 155.0.8059.79
Selenium: 4.51.0

JAVA01 PASS Unknown account rejected
JAVA02 PASS Admin username sign-in
JAVA03 PASS Admin mobile sign-in
JAVA04 PASS Member table search
JAVA05 PASS Six report choices
JAVA06 PASS Trainer profile and scoped navigation
JAVA07 PASS Member profile and scoped navigation
JAVA08 PASS Confirmed logout

Passed: 8   Failed: 0   Total: 8
```

Figure 2. Saved output of the eight executed Java WebDriver checks.

Screenshot of Code and Output (continued)

Selenium WebDriver — code and actual output

Regression code from selenium_suite.py

```
def create_member():
        add_form('members');fill(member_values());save();found=[r for r in data('members') if r['mobile']==phone];assert
len(found)==1;state['member']=found[0]['id'];state['member_name']=tag+' Member';search_for(tag);shot('03-member-management');return
'Created member #'+str(state['member'])

def report_download(name=name):
            nav('reports');click(f'[data-report="{name}"]');file=DOWNLOADS/(name+'-report.pdf');wait.until(lambda d:file.exists() and not
list(DOWNLOADS.glob('*.crdownload')));reader=PdfReader(file);assert len(reader.pages)>=1;assert file.stat().st_size>1000;return
f'{file.stat().st_size} bytes; {len(reader.pages)} PDF pages.'
```

Saved execution metadata


```
{
  "chrome": {
    "run": "20261015-131021",
    "timestamp": "2026-10-15T13:12:20.393016+05:30",
    "base_url": "http://127.0.0.1:8081",
    "browser": "chrome",
    "browser_version": "155.0.8059.79",
    "selenium_version": "4.51.0",
    "tag": "STQA1a198bde",
    "total": 79,
    "passed": 79,
    "failed": 0,
    "cleanup_error": null
  },
  "edge": {
    "run": "20261015-131235",
    "timestamp": "2026-10-15T13:14:01.833772+05:30",
    "base_url": "http://127.0.0.1:8081",
    "browser": "edge",
    "browser_version": "155.0.4283.45",
    "selenium_version": "4.51.0",
    "tag": "STQA0119a856",
    "total": 79,
    "passed": 79,
    "failed": 0,
    "cleanup_error": null
  }
}
```


Figure 3. Actual Python WebDriver regression code and final Chrome/Edge execution metadata.

Screenshot of Code and Output (continued)

Selenium IDE — official runner results

```
{
  "numTotalTests": 30,
  "numPassedTests": 30,
  "numFailedTests": 0,
  "success": true
}
```

1. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Overview
2. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Members
3. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Trainers
4. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Attendance
5. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Equipment
6. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Products
7. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Payment history
8. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Product sales
9. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Workout plans
10. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Progress tracking
11. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Training sessions
12. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Membership plans
13. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin User management
14. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Reports
15. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin Database backup
16. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Trainer My profile
17. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Trainer Assigned members
18. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Trainer My attendance
19. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Trainer Member attendance
20. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Trainer Progress tracking
21. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Trainer Training sessions
22. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Member My profile
23. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Member My attendance
24. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Member My progress
25. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Member My training
26. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Unknown username is rejected
27. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin incorrect password is rejected
28. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Password visibility toggle
29. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Member cannot see administrative links
30. PASS · Running project Gym Management System STQA Running suite Three-role regression Running test Admin sign out

Figure 4. Actual saved official Selenium IDE runner output: 30 passed, zero failed.

(2-3 screenshots of GUI)

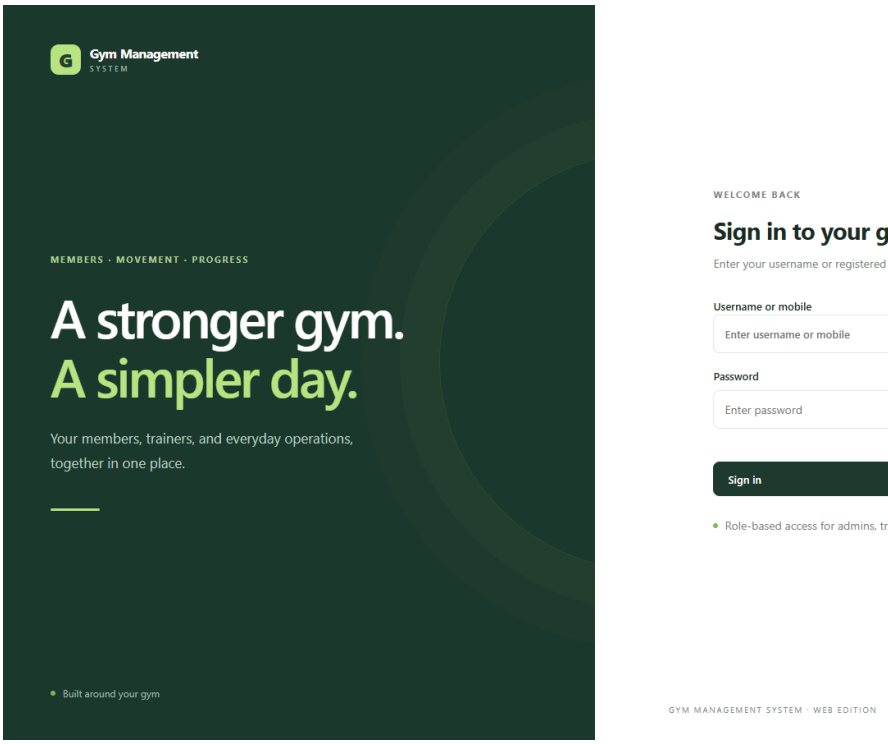


Figure 5. Common login for Admin, Trainer and Member.

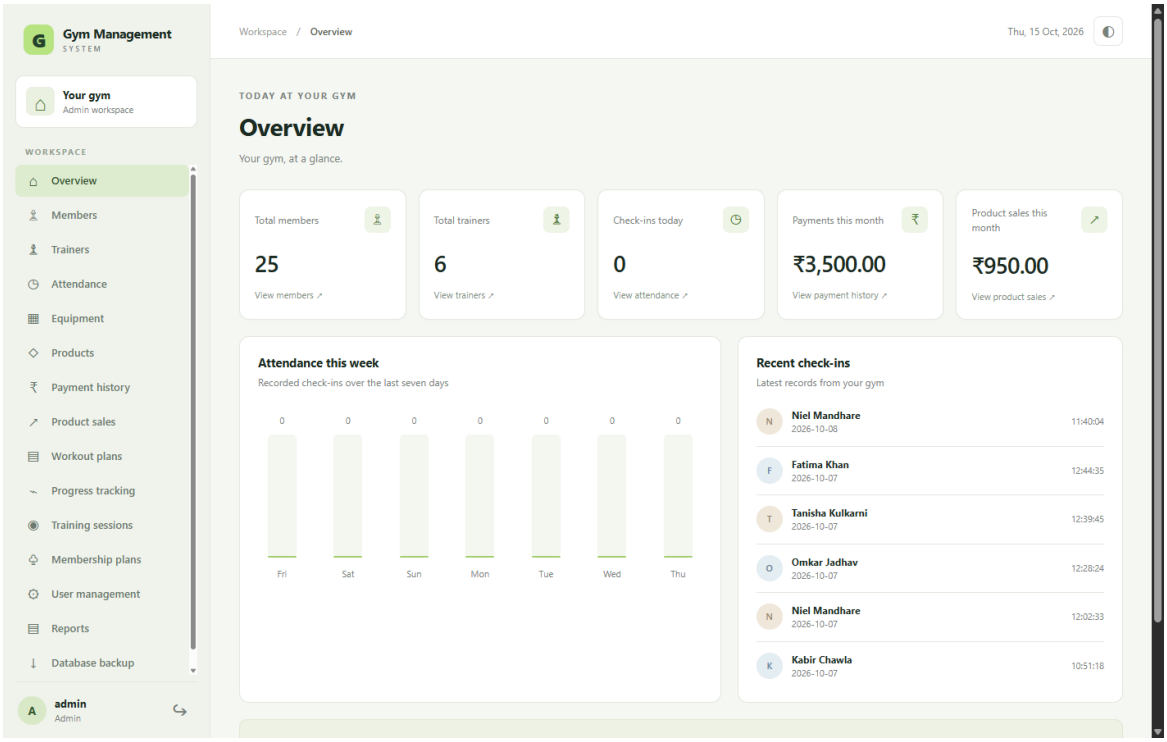


Figure 6. Admin overview with totals from saved gym records.

(2-3 screenshots of GUI) (continued)

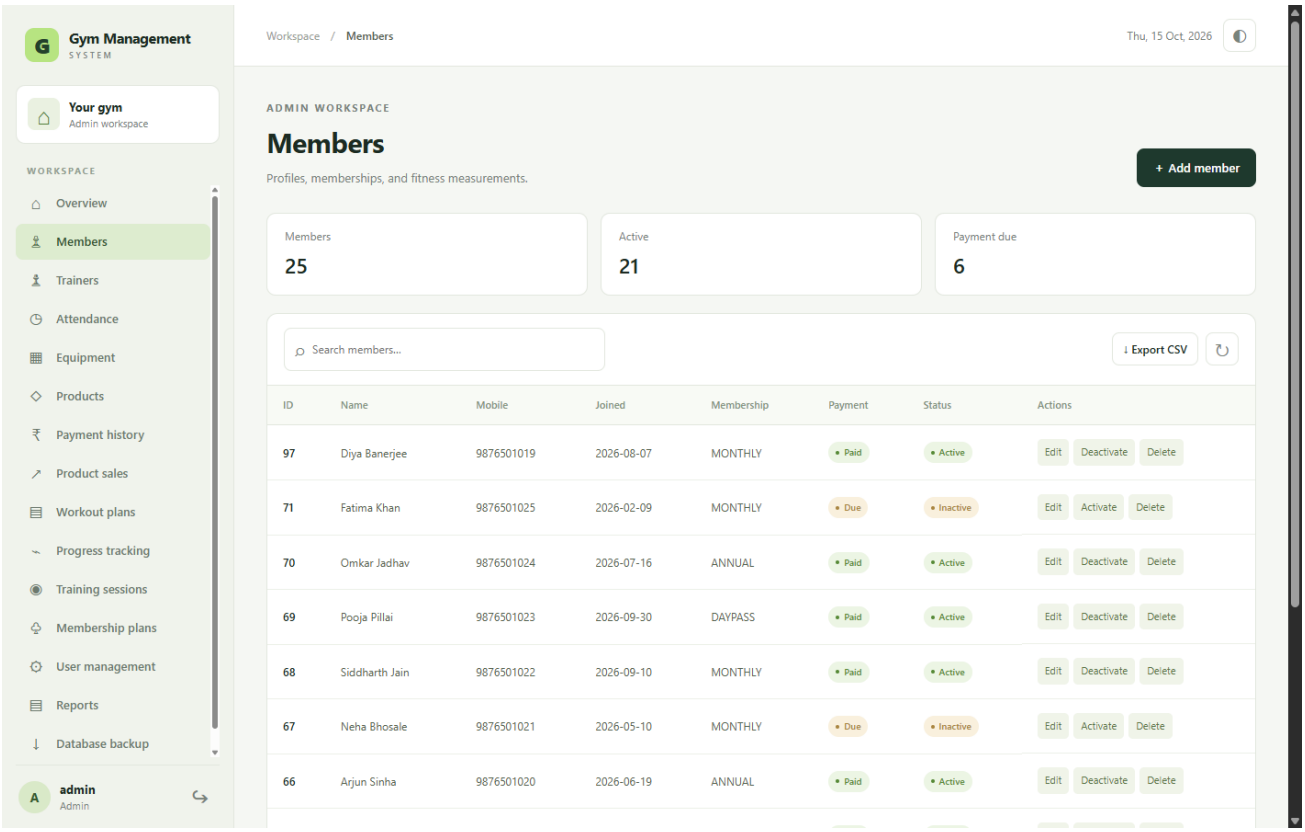


Figure 7. Member management with search, pagination, status and record actions.